MIT 18.06SC - 2.2

Problem 15.1

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$P = A(A^TA)^{-1}A^T$$

$A^TA = I_3$ and $I_3^{-1} = I_3$ so we have $P = AI_3A^T$. $AI_3 = A$ so we have $P = AA^T$.

$$P = AA^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

P is 4×4 and the projection of b onto the column space of A is $= Pb$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$$

Problem 15.2

Suppose P is any matrix such that $P^2 = P$.

Then
$$
\begin{aligned}
(I-P)^2 &= (I-P)(I-P) \\
&= I^2 - IP - PI + P^2 \\
&= I - P - P + P \\
&= I - P
\end{aligned}
$$

Hence ~~proven~~ $(I-P)^2 = (I-P)$

$$
I - P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}
$$

$$
= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

The above matrice ~~sets~~ will project a vector b onto the ~~sp~~ left nullspace of A.